

Massey University

297.101 Statistical Data Science Assignment 2 Questions

Instructions

You need to complete this assignment using RStudio. You are expected to use R mark-down wherever possible.

Please take note that some questions use different quantities for students so that you do the same work as your classmates, but that you get different answers to them because you have slightly different questions or slightly different data.

Once you have completed your work, you will need to upload both the Qmd and HTML files for your assignment to the class Stream site. A drop box will be used to receive the assignments. N.B. If you used the template provided, it forces your Qmd file to be included within the HTML, so that means you only need to upload the HTML.

Individualised solutions files will be made available on Stream once the due date has passed.

1. A New Zealand manufacturer of aircraft components has a local factory as well as an off-shore factory. One particular type of component goes through very strenuous strength tests, and it is not unusual for a high proportion of individual components to fail the test.

The company's quality engineer recently conducted a test on all of the components made in both factories last year. The results from the testing are as follows:

Of the 246 components tested that were made in New Zealand, only 195 passed the strenuous test. Of the 201 components made off-shore that were tested, only 121 passed the same test.

- (a) Construct the full table needed for calculating probabilities, including all row and column totals.

- (b) What is the probability that a randomly selected component is faulty and was made overseas?
- (c) If a randomly selected component is found to be faulty, what is the probability it was made in New Zealand?
- (d) If a randomly selected component is found to be from the New Zealand factory, what is the probability that it was faulty?
- (e) If you could assume the probability that a component was faulty was independent of the probability that it was made in New Zealand, what would the probability be of a randomly selected component being a faulty component produced overseas?
- (f) Justify why you believe the fault rate of parts is or is not independent of where they were manufactured.

This question is worth 7 marks.

2. A telescope contains 3 large mirrors. The time (in years) until a single mirror fails has been investigated by your colleagues. You now know that the probability that a mirror is still fully functional after t years is

$$R(t) = \exp\left(-(t/8.7)^{3.9}\right)$$

- (a) All mirrors must be working to take the most detailed photographs. What is the probability that the telescope can produce these types of pictures for at least 5 years?
- (b) The lowest resolution photographs can be taken as long as at least one mirror is working. What is the probability that these photographs can be taken for at least 7 years?
- (c) The most common photographs the telescope will be taking are of medium resolution. This is possible as long as at least two mirrors are working. What is the probability that this remains possible for at least 4 years?

This question is worth 9 marks.

3. The lifetimes of two types of light bulbs are being compared by a company that must choose one of them for installation in their new office building. The number of hours that 15 light bulbs of each type last is given below:

SampleA=c(970.89, 955.34, 957.06, 966.99, 953.54, 960.19,
953.51, 927.55, 947.53, 965.18, 930.49, 941.41,

953.54, 959.32, 950.3)

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SampleB=c(1939.26, 1922.46, 1929.52, 1966.84, 1936.81, 1921.65,  
1932.73, 1938.36, 1944.67, 1958.51, 1942.52, 1939.89,  
1927.25, 1970.89, 1962.14)
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Note that the second type of bulb costs twice the price of the first.

- (a) Choose your own graphs to compare the distributions of the two samples.
- (b) Explain the differences in your own words and form a recommendation on the type of bulb the company should choose.
- (c) Do the two types of light bulb have a similar distribution? That is, would the same model be appropriate, even if the parameters are different?

This question is worth 7 marks.

4. The data used for this question is synthetic, but is representative of the data collected by Stats NZ as part of the 2018 and 2021 General Social Survey rounds. Each student file is different and is about 99

Your dataset contains survey responses for nearly 2000 synthetic individuals. Many variables have been recoded to suit this assignment, but other variables are still in only the coded format as provided. (Look to the end of the variables for the original coded versions.) Explanations for the values of variables, and how they should be interpreted are in the codebook found on the assignment download page, or in the <https://R-Resources.massey.ac.nz/data/GSS/Codebook.xlsx> file. Feel free to do more cleaning for your assignment, but note that marks are not awarded for doing so.

For each of the following questions, provide the R code, the output, and a sentence describing your conclusion.

- (a) Choose any two demographic variables. Are these two variables independent of one another or associated somehow?
- (b) Use the same approach to see if voting at the last general election is linked to ethnicity. N.B. You need to remove the people who were not eligible to vote.
- (c) Find the proportion of people who are deemed disabled for each gender and ethnicity combination. Tabulate these proportions, the number of survey respondents in each combined group, and the associated standard error for this proportion. Present a graph which displays the proportions and their associated standard errors.

This question is worth $3+3+6=12$ marks.

5. Two brands of digital scales that are purpose designed for measuring small weights are being evaluated for purchase by the university to replace old analog scales. The two brands both display weights to the nearest $1/100$ of a gram and should not be used for weights of more than 150g. Only one brand will be purchased, and the purchase order will be for 150 sets of scales.

The lab staff have obtained 25 sets of scales from both manufacturers and have decided that there is no real reason why either brand should be preferred over the other in terms of their usefulness. Brand A is slightly cheaper than Brand B however so the decision is not a matter of randomly choosing among the brands.

On the advice of the senior lab technician, a benchmark weight of 50g and another of 100g are weighed on all sets of scales to test the consistency of each brand. She believed that these two weights were fairly representative of the weights that will need to be measured in the future.

Your data can be downloaded via Stream, but the template does pull your data for you.

- (a) Accuracy is a measure of biasedness and is calculated as the average of the sampled readings of an instrument minus the target reading that should have been seen. Contrast this with precision, which measures how consistent an instrument is in its readings. Precision is measured by the standard deviation of the readings. What is the accuracy and precision of each brand?
- (b) Form a recommendation on which brand should be purchased.

This question is worth 8 marks.